Dynamic System Modeling and Control
EGR 345 ~ Fall 2026

Instructor: Dr. Ryan Krauss
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Office Hours
Tuesday and Thursday: 10:30 - 11:45am
Other times by appointment and/or virtual (please email me, I am glad to arrange other times to meet)

Course Recordings and FERPA Note
Video and audio recordings of virtual or in-person class sessions may be made available to all students currently enrolled in this section of this course. Your instructor will not share recordings of your class sessions outside of course participants. ***You may not share recordings outside of this course.*** FERPA release from all parties is required for use of recordings for a different class section or future class offering.

Course Description (Official)
An introduction to mathematical modeling of mechanical, thermal, fluid, and electrical systems. Topics include equation formulation, Laplace transform methods, transfer functions, system response and stability, Fourier methods, frequency response, feedback control, control actions, block diagrams, state variable formulation, and computer simulation. Emphasis on mechanical systems. Laboratory. (3-0-3) Offered fall semester. Prerequisites: EGR 214, 226, and MTH 302. Admitted to interdisciplinary or product design and manufacturing engineering major.

Student Learning Objectives (SLOs)

I expect you to

- understand mass/spring/damper systems
    - draw free body diagrams
    - find equations of motion
    - find transfer functions for MSD systems
    - use transfer functions to mathematically model, simulate, and analyze the dynamic response of MSD systems
    - understand the connection between pole locations and the response of a transfer function to various inputs **(extremely important)**
    - know how changing the parameters of a transfer function will affect it's step response
- know how to use Laplace transform analysis and especially partial fraction expansion (PFE) to find the response of a transfer function to any input
- learn to use Bode plots to analyze the frequency response of dynamic systems, find transfer functions, and design feedback control systems
- learn to use block diagrams to model feedback control systems
    - know how to find a closed-loop transfer function based on a block diagram
- learn to design feedback control systems based on open-loop transfer functions using root locus analysis and Bode plots
- become proficient in using Python for dynamic systems and controls purposes:
    - analyze and simulate transfer function models of dynamic systems,
    - design feedback control systems,
    - run experiments using Arduino microcontrollers and serial communication
    - analyze and visualize experimental data
- develop and conduct appropriate experiments for dynamic systems, analyze and interpret experimental data, and use engineering judgment to draw conclusions from data

Textbook

Required: ***System Dynamics - Katsuhiko Ogata***

- Third edition or higher recommended
- Second edition mostly acceptable

Course Policies

- Students are expected to conduct themselves in a professional manner:
    - respect the teacher
    - respect one another
    - direct all questions to the teacher
    - be on time to class
    - turn cell phones off (unless there is a family medical emergency)
- Academic Honesty will be taken very seriously
    - All suspected violations will be referred to OSCCR
- Attendance is expected (in person or virtual)
    - If you are sick or choose not to attend lecture face-to-face, you are still expected to participate in class activities virtually via Blackboard collaborate.
    - If you are too sick to participate in class virtually, contact me to make other arrangements.

Special Accommodations

If you are in need of accommodations due to a learning, physical, or other disability you must present a memo to me from Disability Support Resources (DSR), indicating the existence of a disability and the suggested reasonable accommodations. If you have not already done so, please contact the Disability Support Resources office (4015 JHZ) by calling 331-2490 or email to dsrgvsu@gvsu.edu. Please note that I cannot provide accommodations based upon disability until I have received a copy of the DSR issued memo. All discussions will remain confidential.

Furthermore, if you have a physical disability and think you will need assistance evacuating this classroom and/or building in an emergency situation, please come talk to me so that we can work out a plan together.

Grading

Learning Activities and Homework:	15%
Labs 1-6:	15%
Mini Projects	10%
Quizzes:	10%
Exam 1:	10%
Exam 2	10%
Practical Lab Exam (individual):	10%
Final Exam:	20%

Final Exam Time

Tuesday, December 15, 8:00 am - 9:50 am
(from registrar's website; based on class start time of Tuesday at 9:00 am)